

- 1) Let's assume that the no. of commuters approving HOK lane (X_i) be distributed according to a binomial distribution

$$Y_i | \theta \sim \text{Binomial}(m_i, \theta)$$

where $m_i =$ No. of commuters approving X_i ✓
+ No. of commuters disapproving X_i ✓

- a) The log-likelihood is given by ^{for a given county}

$$L(\theta | \mathbf{X}) = \log \left[\binom{m_i}{X_i} \theta^{X_i} (1-\theta)^{m_i - X_i} \right]$$

$$= X_i \log(\theta) + (m_i - X_i) \log(1-\theta) + \text{constants}$$

$$\frac{\partial L}{\partial \theta} = \frac{X_i}{\theta} - \frac{m_i - X_i}{1-\theta}$$

$$\frac{\partial^2 L}{\partial \theta^2} = -\frac{X_i}{\theta^2} - \frac{m_i - X_i}{(1-\theta)^2}$$

$$F(\theta) = -E_{Y|X} \left(\frac{\partial^2 L}{\partial \theta^2} \right)$$

$$= \frac{m_i}{\theta(1-\theta)}$$

Jeffreys' prior distribution is given by

$$\pi(\theta) \propto \sqrt{I(\theta)}$$

let

$$\pi(\theta) = \frac{1}{\sqrt{\theta(1-\theta)}}$$

The posterior distribution of $\theta | Y_i$ is given by

$$p(\theta | Y_i) = p(Y_i | \theta) \pi(\theta)$$

~~$$\propto \theta^{Y_i + \frac{1}{2} - 1} (1-\theta)^{m_i - Y_i + \frac{1}{2} - 1}$$~~

$$\propto \theta^{Y_i} (1-\theta)^{m_i - Y_i} \times \theta^{-1/2} (1-\theta)^{-1/2}$$

$$= \theta^{Y_i + \frac{1}{2} - 1} (1-\theta)^{m_i - Y_i + \frac{1}{2} - 1}$$

This is the kernel density of $\text{Beta}(Y_i + \frac{1}{2}, m_i - Y_i + \frac{1}{2})$

$$\theta | Y_i \sim \text{Beta}(Y_i + \frac{1}{2}, m_i - Y_i + \frac{1}{2})$$

We can find a 95% posterior credible interval using quantiles of Beta in R

b) let $m = \text{mean of } \hat{\beta}_i$
 where $i = 1, \dots, n > 10$
 & $V = \text{Variance of } \hat{\beta}_i$

Mean & Variance of Beta (a, b)
 are given by $\frac{a}{a+b}$ & $\frac{ab}{(a+b+1)(a+b)^2}$
 respectively

$$m = \frac{a}{a+b} \quad \text{--- (1)}$$

$$V = \frac{ab}{(a+b+1)(a+b)^2} \quad \text{--- (2)}$$

Solving (1) & (2) we get

$$a = \frac{m^2(1-m)}{V} - m$$

&

$$b = \frac{m(1-m)^2}{V} - (1-m)$$

c) $\theta \sim \text{Beta}(a, b)$

The posterior distribution $\theta | Y_i$ pdf is given by

$$p(\theta | Y_i) = p(Y_i | \theta) \pi(\theta)$$

$$\propto \theta^{Y_i} (1-\theta)^{n_i - Y_i} \theta^{a-1} (1-\theta)^{b-1}$$

$$= \theta^{a+Y_i-1} (1-\theta)^{n_i+Y_i+b-1}$$

$$\theta_i | Y_i \sim \text{Beta}(a+Y_i, n_i+Y_i+b)$$

We can find the ^{posterior} credible sets using quantiles in R

2)

$$X_i | \mu \sim N(\mu, \sigma_i^2)$$

$$i \in \{1, \dots, n\}$$

$\sigma_i \rightarrow \text{known}$

$$\pi(\mu) = 1 \quad \forall \mu$$

a) MAP estimator of μ is given by

$$\hat{\mu}_{\text{MAP}} = \underset{\mu}{\operatorname{argmax}} (\log(f(X_i | \mu)) + \log(\pi(\mu)))$$

$$= \underset{\mu}{\operatorname{argmax}} \left(\log \left[\prod_{i=1}^n (\sigma_i^2)^{-1/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma_i} \right)^2 \right] \right] + \log(1) + \log(\text{constants}) \right]$$

$$= \underset{\mu}{\operatorname{argmax}} \left(-\frac{1}{2} \sum_{i=1}^n \log(\sigma_i^2) - \frac{1}{2} \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma_i^2} \right)$$

$$= \underset{\mu}{\operatorname{argmax}} \left(\underbrace{-\frac{1}{2} \left[\sum_{i=1}^n \left(\frac{X_i}{\sigma_i} \right)^2 - 2\mu \sum_{i=1}^n \frac{X_i}{\sigma_i^2} + \mu^2 \sum_{i=1}^n \frac{1}{\sigma_i^2} \right]}_{\rightarrow A} \right)$$

~~2A~~

taking $\partial A / \partial \mu = 0$

$$\rightarrow \frac{\partial A}{\partial \mu} = \frac{-1}{2} \left(-2 \sum_{i=1}^n \frac{X_i}{\sigma_i^2} + 2\mu \sum_{i=1}^n \frac{1}{\sigma_i^2} \right) = 0$$

$$\rightarrow \mu = \frac{\sum_{i=1}^n \frac{X_i}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}$$

$$\frac{\partial^2 A}{\partial \mu^2} = - \sum_{i=1}^n \frac{1}{\sigma_i^2} < 0$$

$$\text{So, } \hat{\mu}_{MAP} = \frac{\sum_{i=1}^n \frac{y_i}{\sigma_i^2}}{\sum_{i=1}^n \frac{1}{\sigma_i^2}}$$

b) $n = 3, y_1 = 12, y_2 = 10, y_3 = 22$
 $\sigma_1 = \sigma_2 = 3, \sigma_3 = 10$

$$\begin{aligned} \hat{\mu}_{MAP} &= \frac{\frac{12}{9} + \frac{10}{9} + \frac{22}{100}}{\frac{1}{9} + \frac{1}{9} + \frac{1}{100}} \\ &= \frac{22 \left(\frac{1}{9} + \frac{1}{100} \right)}{\left(\frac{2}{9} + \frac{1}{100} \right)} = \frac{22 \left(\frac{109}{900} \right)}{\left(\frac{209}{900} \right)} \\ &= \frac{22 \times 109}{209} \approx 11.47 \end{aligned}$$

c) $f(y_1, \dots, y_n | \mu) = \frac{1}{(2\pi)^{n/2}} \prod_{i=1}^n (\sigma_i^2)^{-1/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{y_i - \mu}{\sigma_i} \right)^2 \right]$

$$p(\mu | y_1, \dots, y_n) \propto \exp \left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{y_i - \mu}{\sigma_i} \right)^2 \right] \times (1)$$

So, Posterior mean of μ is given by

$$E(\mu | y_1, \dots, y_n) = \frac{\int_{-\infty}^{\infty} \mu \exp \left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{y_i - \mu}{\sigma_i} \right)^2 \right] d\mu}{\int_{-\infty}^{\infty} \exp \left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{y_i - \mu}{\sigma_i} \right)^2 \right] d\mu}$$

3)

$$Y_i | \sigma_i^2 \sim N(0, \sigma_i^2)$$

$$i = 1, \dots, n$$

$$\sigma_i^2 | b \sim \text{IG}(a, b)$$

$$b \sim \text{Gamma}(1, 1)$$

a) Full conditional posterior distribution of σ_i^2 is given by

$$p(\sigma_i^2 | \text{rest}) = p(Y_1, \dots, Y_n | \sigma_1^2, \dots, \sigma_n^2) p(b | b) p(b)$$

$$\propto \prod_{i=1}^n (\sigma_i^2)^{-1/2} \exp\left[-\frac{1}{2} \sum_{i=1}^n \left(\frac{Y_i}{\sigma_i^2}\right)^2\right] \times (\sigma_i^2)^{-a-1} \exp\left[-\frac{b}{\sigma_i^2}\right]$$

$$\propto b^{1-1} \exp(-b) (\sigma_i^2)^{-a-\frac{1}{2}-1} \exp\left[-\frac{1}{\sigma_i^2} \left(b + \frac{Y_i^2}{2}\right)\right]$$

$$\text{So, } \sigma_i^2 | \text{rest} \sim \text{IG}\left(a + \frac{1}{2}, b + \frac{Y_i^2}{2}\right)$$

Full conditional posterior distribution of b is given by

$$\cancel{p(b | \text{rest}) = p(\sigma_1^2 | \text{rest}) \dots p(\sigma_n^2 | \text{rest}) p(b)}$$

$$p(b | \text{rest}) = p(\sigma_1^2, \dots, \sigma_n^2 | \text{rest}) p(b)$$

$$\propto \prod_{i=1}^n (\sigma_i^2)^{-a-\frac{1}{2}-1} \exp\left[-\frac{1}{\sigma_i^2} \left(b + \frac{Y_i^2}{2}\right)\right]$$

$$\propto b^{1-1} \exp(-b)$$

$$\propto b^{n-1} \exp \left[-b \left(1 + \sum_{i=1}^n \frac{1}{\sigma_i^2} \right) \right]$$

$$b | \text{rest} \sim \text{Gamma} \left(1, 1 + \sum_{i=1}^n \frac{1}{\sigma_i^2} \right)$$

b) Gibbs sampling algorithm

1) Set $\theta^{(0)} = (\sigma_1^2, \dots, \sigma_n^2, b^{(0)})$

2) For iteration t

Draw $\sigma_i^2 \sim \text{IG} \left(q + \frac{1}{2}, b^{(t-1)} + \frac{y_i^2}{2} \right)$

$$b^{(t)} \sim \text{Gamma} \left(1, 1 + \sum_{i=1}^n \frac{1}{\sigma_i^2} \right)$$

3) Repeat step 2 until the desired no. samples are drawn after convergence.

4) $Y_i | \theta_i \sim \text{Binomial}(n_i, \theta_i)$ $i = 1, \dots, 10$
 $\theta_i | m \sim \text{Beta}[\exp(m)q_i, \exp(m)(1-q_i)]$

where $q_i \in (0, 1)$

$m \sim N(0, 10)$

a) This is a reasonable prior for θ_i as Beta is a conjugate prior for Binomial distribution

b)

c) Full conditional posterior θ_i is given by

$$p(\theta_i | \text{rest}) = p(Y_1, \dots, Y_n | \theta_1, \dots, \theta_n) p(\theta_i | m) / p(m)$$

$$\propto \prod_{i=1}^n \theta_i^{y_i} (1-\theta_i)^{n_i-y_i} \theta_i^{\exp(m)q_i-1} (1-\theta_i)^{\exp(m)(1-q_i)-1}$$

$$\times \exp\left(-\frac{m^2}{20}\right)$$

$$\propto \theta_i^{\exp(m)q_i+y_i-1} (1-\theta_i)^{n_i-y_i+\exp(m)(1-q_i)-1}$$

So, $\theta_i | \text{rest} \sim \text{Beta}(Y_i + \exp(m)q_i, n_i - Y_i + \exp(m)(1-q_i))$

d) we use Gibbs sampling to solve this problem

Algorithm:

1) Set $\Theta^{(0)} = (\theta_1^{(0)}, \dots, \theta_{p-1}^{(0)}, m^{(0)})$

2) At t^{th} iteration

draw $\theta_i^{(t)} \sim \text{Beta}(\gamma_i + \exp(m)q_i, m - \gamma_i + \exp(m)(1 - q_i))$

draw $m^{(t)} \sim N(q|0)$

3) Repeat step (2) until desired no. of samples are drawn after convergence

4) $\text{Mean}_{\text{Kth}} = \frac{1}{K} \sum_{t=1}^K \Theta^{(t)}$

& for 95% CI

like like 2.5% th quantile as the lower bound & 97.5% th quantile as the upper bound

5) Y_i i.i.d Laplace (μ, σ)
 $i = 1, \dots, m; \theta = (\mu, \sigma)$

$$\sigma \sim U(0, 10^5)$$

$$\pi(\mu) = 1 \quad \mu \in (-\infty, \infty)$$

$$\begin{aligned} a) p(\theta | Y) &\propto p(Y | \mu, \sigma) p(\mu) p(\sigma) \\ &= \frac{1}{2\sigma} \exp\left(-\sum_{i=1}^m \frac{|y_i - \mu|}{\sigma}\right) \frac{1}{10^5} \mathbb{I}(\sigma \in [0, 10^5]) \\ &= \frac{1}{2 \times 10^5 \sigma} \exp\left(-\sum_{i=1}^m \frac{|y_i - \mu|}{\sigma}\right) \mathbb{I}(\sigma \in [0, 10^5]) \end{aligned}$$

$$b) E(\mu | Y) = \frac{\int_0^{10^5} \int_{-\infty}^{\infty} \frac{\mu}{2 \times 10^5 \sigma} \exp\left(-\sum_{i=1}^m \frac{|y_i - \mu|}{\sigma}\right) d\mu d\sigma}{NC}$$

$$E(\sigma | Y) = \frac{\int_0^{10^5} \int_{-\infty}^{\infty} \frac{1}{2 \times 10^5 \sigma} \exp\left(-\sum_{i=1}^m \frac{|y_i - \mu|}{\sigma}\right) d\mu d\sigma}{NC}$$

$$\text{where } NC = \int_{-\infty}^{\infty} \int_0^{\infty} \frac{1}{2\sigma} \exp\left(-\sum_{i=1}^m \frac{|y_i - \mu|}{\sigma}\right) d\sigma d\mu$$

$$c) f(\hat{Y} | Y) = \int p(\theta | Y) f(Y | \theta) d\theta$$

$$\begin{aligned} &\int_0^{10^5} \int_{-\infty}^{\infty} \frac{1}{2\sigma} \exp\left(-\frac{|y_{m+1} - \mu|}{\sigma}\right) \times \frac{1}{2 \times 10^5 \sigma} \exp\left(-\sum_{i=1}^m \frac{|y_i - \mu|}{\sigma}\right) d\mu d\sigma \\ &= \int_0^{10^5} \int_{-\infty}^{\infty} \frac{1}{4 \times 10^5 \sigma^2} \exp\left(-\sum_{i=1}^{m+1} \frac{|y_i - \mu|}{\sigma}\right) d\mu d\sigma \end{aligned}$$